

Experiment no-02

3. Write program demonstrates how to perform morphological analysis using the NLTK library in Python.

PROGRAM:

```
import nltk
from nltk.tokenize import word_tokenize
from nltk.stem import PorterStemmer

# Download required resources (Run only once)
nltk.download('punkt')
nltk.download('punkt_tab')

ps = PorterStemmer()

text = input("Enter a sentence: ")

words = word_tokenize(text)

print("\nOriginal Word\tStemmed Word")
print("-----")

for word in words:
    print(f"{word}\t\t{ps.stem(word)}")
```

OUTPUT:

```
... [nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt_tab.zip.
Enter a sentence: i love chocolates

Original Word  Stemmed Word
-----
i             i
love          love
chocolates    chocol
```